

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

RADAR AND MICROWAVE COMMUNICATION**(E)**

(Time : 3 hours)

(Maximum marks:100)

PART – A(Answer *all* questions in one or two sentences. Each question carries 2 marks)**I**

1. What are the basic functions of a radar?
2. State the difference between pulse width and pulse repetition time
3. What is the need for delay line cancellers
4. List the modes of operation for rectangular waveguides
5. What is meant by fading

2x5=10

PART – B(Answer any *five* of the following questions. Each question carries 6 marks)**II**

1. What are the factors to be considered practically in fixing the frequency of a radar?
2. What is meant by radar tracking? Describe different types?
3. With block diagram, explain MTI signal processor?
4. Compare the working principle of IMPATT diodes and TRAPATT diodes
5. Briefly explain about circulators and isolators used in microwave?
6. Describe the principle of a digital microwave link?
7. Draw the block diagram of a terminal station microwave receiver and explain?

5x6=30

PART – C(Answer *one* full question from each module. Each question carries 15 marks)**MODULE – I**

- III**
1. Derive radar range equation?

9

2. Describe the factors governing the relation between the radar cross section of a target and its true cross section?	6
OR	
IV 1. Describe the noise and its effect in radar range?	8
2. A radar transmitter has a peak pulse power of 200kW, a PRF of 1500pps and a pulse width of 0.8μs. Calculate the maximum range, the duty cycle and average transmitted power?	7
MODULE – II	
V 1. Explain the operation of MTI radar using power amplifier output?	10
2. What are the advantages, applications and limitations CW Doppler radar?	5
OR	
VI 1. Briefly explain different types of radar displays?	7
2. What is Doppler effect? Explain one type of radar working under this principle	8
MODULE – III	
VII 1. With necessary diagram, explain the working of Reflex Klystron	10
2. List the applications of Gunn diode	5
OR	
VIII 1. Explain the principle of operation of a Magnetron Oscillator?	8
2. Describe the working principle of Microwave FETs?	7
MODULE – IV	
IX 1. Explain the working principle of a microwave link with block diagram	10
2. Write short notes on polarization diversity	5
OR	
X 1. Draw and explain the block diagram of a basic microwave communication system	7
2. Explain diversity method of protection switching?	8